

Local recurrence of uveal melanoma after proton beam therapy

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Proton beam therapy in uveal melanoma shows a local recurrence rate ranging 2–8.4 per cent after five years. Risk factors for failure of local control after proton beam therapy include large tumour diameter, small tumour volume and safety margins of less than 1 mm^{3,1}

Currently, enucleation is still the most used treatment for recurrent melanoma. The treatment approach depends on the extent and location of the recurrence; however, it remains a case-by-case decision.

To date, there are few published studies comparing the genetic basis of local recurrence with primary uveal melanoma tumour. The clinical and genetic characteristics of a patient with local recurrence of a peripapillary

uveal melanoma treated with proton beam therapy are presented.

Clinical case

A 65-year-old male was referred to the Adult Intraocular Tumours Unit at the Ophthalmology Service of Santiago de Compostela due to myodesopsias and progressive visual loss in his right eye. Ophthalmological examination showed a nasal-superior peripapillary choroidal mass (270° in contact with the optic nerve) of nodular morphology and abundant orange pigment on the surface.

The patient also presented with an inferior exudative retinal detachment with separation between the pigmentary epithelium and the neurosensorial retina, including subretinal fluid in its interior, and complete loss of the foveal profile. Ocular ultrasound showed features compatible with a choroidal melanoma of base size 11.5 mm and height 3.75 mm. A clinical extension study that ruled out metastatic spread was performed. Given the location and size of the primary tumour, proton beam treatment was proposed (Figure 1).

Six months later, a new nodular choroidal mass, centred on the posterior pole that extended to the lower peripapillary quadrant, was detected and identified as a new extension of the first primary tumour mass. Still, there were no signs of metastatic disease. The visual acuity on the first visit was 6/15 and decreased to fingers count in the last examination (Figure 2).

Given the new findings, enucleation of the right eye was proposed. The anatomopathological study of both masses resulted in epithelioid-type uveal melanoma.

After enucleation, a genetic study was conducted to detect gains or losses in chromosomes 1p, 3, 8 and 6 in both masses using a Salsa multiplex ligation-dependent probe amplification (MLPA) probemix P027 MRC-Holland assay kit. The biopsy of the

primary tumour, previously treated by proton beam therapy showed a monosomy of chromosome 3, loss of the short arm of chromosome 1(1p) and gain of the 8q region. Biopsy of the recurrence mass showed monosomy of chromosome 3, gain of 8q and 6p regions and loss of 6q.

Seven months after enucleation, the patient died due to liver metastases.

Discussion

Uveal melanoma tumour recurrence correlates with an increased risk of metastasis and reduced survival rate.¹ The genetic analysis in the present study resulted in both the primary tumour and the recurrence mass being designated as 'high risk', with a high probability of developing metastasis. However, each tumour mass showed its own genetic profile with different chromosomal aberrations; thus, in the recurrence, chromosome 1 short arm was unaltered, and gain of 6p and loss of 6q was detected.

Over the past decade, tumoral cytogenetic profile has been revealed to be of great importance in uveal melanoma prognosis. Indeed, monosomy of chromosome 3, chromosome 8 long arm gains, and the loss of 1p and 6q regions are related to a poor disease prognosis. In fact, the loss of one copy of chromosome 3 seems to be an early event in uveal melanoma and may lead to increased genomic instability, especially in the induction of isochromosome formation.² On the other hand, gains at the short arm of chromosome 6 are related to a favourable outcome of uveal melanoma.^{3,4} Although monosomy of chromosome 3 and chromosome 6p gains are generally mutually exclusive, and occur in just four per cent of cases of uveal melanoma,⁵ the relapsing mass in the present case showed this genetic association.⁶

While it is known that recurrence is associated with a higher risk of mortality, there are currently no studies describing the genetic signature of local uveal melanoma

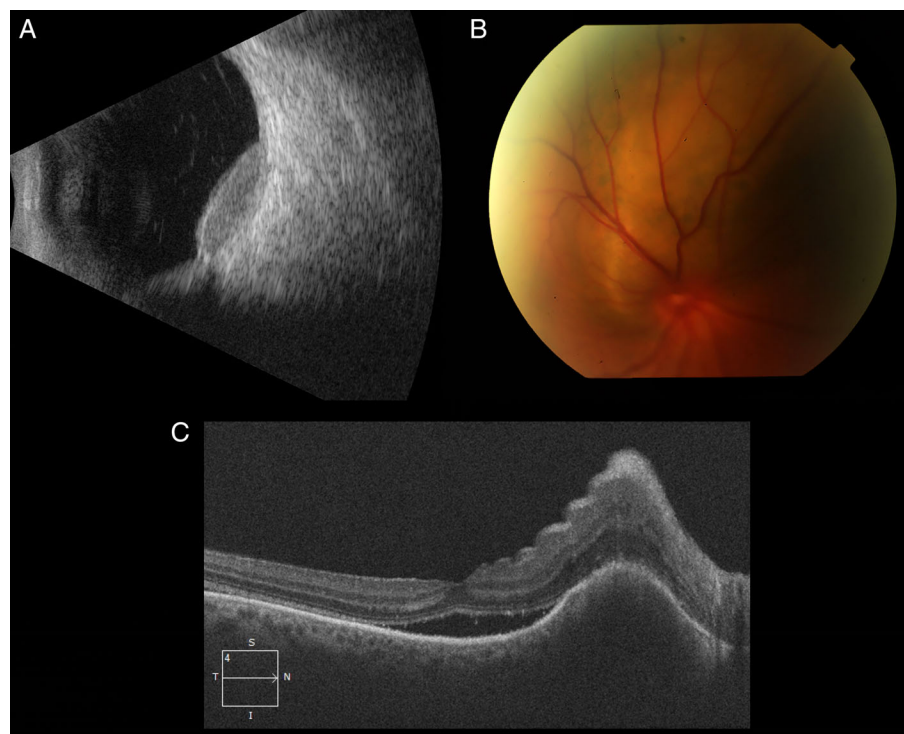


Figure 1. A: Ocular echography and B: retinography of choroidal melanoma before proton beam treatment. C: Image obtained by optical coherence tomography. Serous retinal detachment of the macula.

recurrences. In this case, a common feature is established in chromosomes 3 and 8 alterations, but there is a change in chromosomes 1 and 6. These differences could be explained by the existence of genetic heterogeneity in the primary tumour, and the growth of one clone of cells in the secondary mass extension. Alternatively, another explanation could be the transformation of the primary tumour with the sum of new chromosome 6 aberrations and the disappearance of 1p alteration in a new clone.

Unfortunately, a sample from the primary tumour without proton beam treatment was not accessible in the present case. However, this limitation may not be so important, as the results of other studies, such as those from Krishna et al.⁵ and Gold et al.,⁷ showed that proton beam therapy and brachytherapy do not modify the tumoral genetic conditions.

There appear to be no published studies comparing the genetic basis of local recurrence with the primary tumour in uveal melanoma. In this case, the clinical and genetic characteristics of a patient with local recurrence of a peripapillary uveal melanoma previously treated with proton beam radiation are presented. These results suggest that recurrences in uveal melanoma do not necessarily have the same genetic profile as the primary tumour, and therefore may not have the same disease prognosis.

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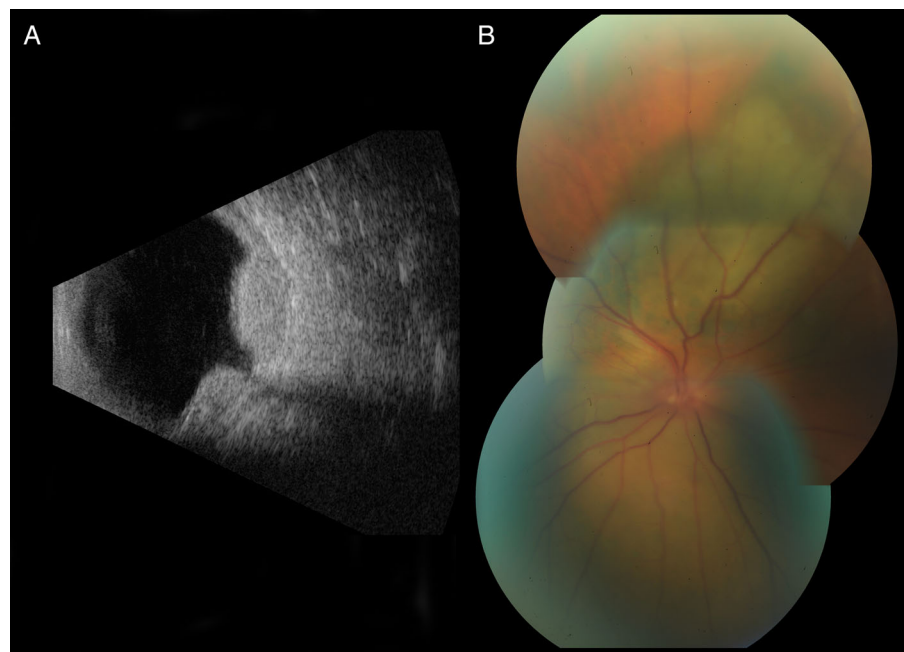


Figure 2. Six months after proton beam treatment. Tumour recurrence demonstrated as tumour growth in A: ocular echography and B: fundus image.